Problem 7

A square D contains two regions D_1 and D_2 with areas S_1 and S_2. A point is thrown into D and can fall into any part of the square with equal probability. Let event A_k denote the point falling into region D_k ($k = 1, 2$). Are events A_1 and A_2 equally likely?

Solution

We need to determine whether the probabilities of events A_1 and A_2 are equal.

Step 1: Setup and Definitions

Given Information

- Square D with some total area S (the area of square D)
- Region $D_1 \subseteq D$ with area S_1
- Region $D_2 \subseteq D$ with area S_2
- A point is thrown uniformly at random into D
- Event A_1 = "point falls in region D_1"
- Event A_2 = "point falls in region D_2"

Step 2: Probability Calculation

Since the point falls into any part of the square with equal probability (uniform distribution), the probability of the point landing in a region is proportional to the area of that region.

Probability Formula

For a uniform distribution over the square D with area S:

$$P(A_1) = \frac{\text{Area of } D_1}{\text{Area of } D} = \frac{S_1}{S}$$

$$P(A_2) = \frac{\text{Area of } D_2}{\text{Area of } D} = \frac{S_2}{S}$$

Step 3: When Are Events Equally Likely?

Definition

Two events are **equally likely** (equiprobable) if they have the same probability:

$$P(A_1) = P(A_2)$$

Condition for Equal Likelihood

$$P(A_1) = P(A_2) \iff \frac{S_1}{S} = \frac{S_2}{S} \iff S_1 = S_2$$

Step 4: Answer to the Question

$$\boxed{\text{Events } A_1 \text{ and } A_2 \text{ are equally likely if and only if } S_1 = S_2}$$

In general: Without additional information about the relationship between S_1 and S_2, we **cannot conclude** that A_1 and A_2 are equally likely.

Step 5: Important Distinction

Equally Likely vs. Equally Possible

It's important to distinguish between:

- **Equally possible** (mogući): Both events can occur; neither is impossible
- **Equally likely** (jednako verovatni): Both events have the same probability

Answer:

- Events A_1 and A_2 are both **equally possible** (assuming $S_1, S_2 > 0$)
- Events A_1 and A_2 are **equally likely** if and only if $S_1 = S_2$

Step 6: Special Cases

Case 1: Disjoint Regions ($D_1 \cap D_2 = \emptyset$)

If D_1 and D_2 don't overlap:

$$P(A_1) = \frac{S_1}{S}, \quad P(A_2) = \frac{S_2}{S}$$

They are equally likely if and only if $S_1 = S_2$.

Case 2: Overlapping Regions ($D_1 \cap D_2 \neq \emptyset$)

If D_1 and D_2 overlap, the events are not mutually exclusive. A point could fall in both regions simultaneously. The probabilities are still:

$$P(A_1) = \frac{S_1}{S}, \quad P(A_2) = \frac{S_2}{S}$$

They are equally likely if and only if $S_1 = S_2$.

Case 3: One Region Contained in Another

If $D_1 \subset D_2$, then $A_1 \subset A_2$, and:

$$P(A_1) = \frac{S_1}{S} < \frac{S_2}{S} = P(A_2)$$

They are not equally likely (unless $S_1 = S_2$, which would mean $D_1 = D_2$).

Step 7: Example

Example 1: Equal Areas

Let D be a square with side length 10 (area $S = 100$).

- D_1 is a circle with radius 2 (area $S_1 = 4\pi \approx 12.57$)
- D_2 is a square with side $2\sqrt{\pi}$ (area $S_2 = 4\pi \approx 12.57$)

Since $S_1 = S_2$:

$$P(A_1) = P(A_2) = \frac{4\pi}{100} \approx 0.126$$

Events A_1 and A_2 are equally likely.

Example 2: Different Areas

Let D be a unit square (area $S = 1$).

- D_1 is the left half (area $S_1 = 0.5$)
- D_2 is the top quarter (area $S_2 = 0.25$)

Then:

$$P(A_1) = 0.5 \neq 0.25 = P(A_2)$$

Events A_1 and A_2 are NOT equally likely.

Final Answer

$$\boxed{\text{Events } A_1 \text{ and } A_2 \text{ are equally likely if and only if } S_1 = S_2}$$

Detailed Answer:

- The probability of event A_k is $P(A_k) = \frac{S_k}{S}$, where S is the area of square D
- Events A_1 and A_2 are equally likely $\iff P(A_1) = P(A_2) \iff S_1 = S_2$
- **Without additional information about the relationship between S_1 and S_2, we cannot conclude that the events are equally likely**

Key Principle: In geometric probability with uniform distribution, the probability is proportional to area (or length in 1D, volume in 3D).

Important Note: The problem asks if the events are "jednako mogući" (equally possible/likely). Both events are *possible* as long as $S_1, S_2 > 0$, but they are *equally likely* only when $S_1 = S_2$.